The K1-invariants of a subquotient as a fibre product of truncations

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned publisher source of this unit; no statement or proof step was rewritten, reordered or removed. The counted statement is the labelled corollary environment at source lines 3588-3596 and the counted proof is the proof environment at source lines 3598-3621.

Editorial note. The preamble of the delivered fragment is generated for this unit: the publisher class is replaced by article plus the ams packages and the author's own package, macro and theorem-declaration lines are retained in source order together with their continuation lines; publisher meta commands are removed.

Editorial note. The two results invoked by the counted text are declared as non-load-bearing external pointers with their printed numbers rather than restated.

Allowed standard prerequisites (editorial).

- Identification of the natural map between coinvariants. The author's theorem (printed number 6.18). Identifies the kernel computation with a map between explicitly described modules, which is the object of the proof.
- The socle of a subquotient. The author's earlier corollary (printed number 6.15). Supplies the description of the socle used in the reduction to the case treated by the preceding results.
- Truncations of the fixed module and their fibre products. The author's construction of the modules attached to a parameter in the same article. Provide the modules whose fibre product appears in the statement and the filtration with respect to which the proof argues.

Source reference. Original theorem 6.18: sources/jep-2026-346-on-the-constituents-of-the-mod-cohomology-of-shimura/pinned-original-source.tex, source lines 2720–2726. Prerequisite pointer invoked by the counted proof: it identifies the natural map between the coinvariant spaces with a map between the displayed modules.

Source reference. Original corollary 6.15: sources/jep-2026-346-on-the-constituents-of-the-mod-cohomology-of-shimura/pinned-original-source.tex, source lines 2586–2595. Prerequisite pointer invoked by the counted proof for the description of the socle of a subquotient.

Corollary 0.0.1. Assume that $\overline{\rho}$ is $\max\{9, 2f+3\}$-generic. Suppose $\pi' = \pi_1'/\pi_1$ is any nonzero subquotient, where $\pi_1 \subsetneq \pi_1' \subset \pi$. Let $i_0 \overset{\text{def}}{=} i_0(\pi_1)$, $i_0' \overset{\text{def}}{=} i_0(\pi_1')$, so $-1 \leq i_0 < i_0' \leq f$. Then

$$\pi'^{K_1} \cong D_0(\overline{\rho}^{\mathrm{ss}})_{i_0+1} \oplus_{D_0(\overline{\rho})_{i_0+1}} (D_0(\overline{\rho})_{\leq i_0'}/D_0(\overline{\rho})_{\leq i_0}).$$

Proof. Note that π'^{K_1} is the kernel of the natural map $(\pi/\pi_1)^{K_1} \to (\pi/\pi_1')^{K_1}$. Let us write again

$$D_{i_0} \overset{\text{def}}{=} D_0(\overline{\rho}^{\mathrm{ss}})_{i_0+1} \oplus_{D_0(\overline{\rho})_{i_0+1}} (D_0(\overline{\rho})/D_0(\overline{\rho})_{\leq i_0}).$$

By Theorem 6.18 the above natural map is identified with a map

$$\theta : D_{i_0} \to D_{i_0'}.$$

Let $\theta_0 \overset{\text{def}}{=} \theta|_{D_0(\overline{\rho})/D_0(\overline{\rho})_{\leq i_0}}$. We have

$$\mathrm{soc}_{\mathrm{GL}_2(\mathcal{O}_K)}(\pi/\pi_1') \subset \{\sigma \in W(\overline{\rho}^{\mathrm{ss}}) : \ell(\sigma) \geq i_0' + 1\}$$

by Corollary 6.15, and $\mathrm{JH}(D_0(\overline{\rho}^{\mathrm{ss}})_{i_0+1}) \cap W(\overline{\rho}^{\mathrm{ss}})$ is disjoint from that set since $i_0 < i_0'$, so $D_0(\overline{\rho}^{\mathrm{ss}})_{i_0+1} \subset \ker(\theta)$ and hence

$$\ker(\theta) = D_0(\overline{\rho}^{\mathrm{ss}})_{i_0+1} \oplus_{D_0(\overline{\rho})_{i_0+1}} \ker(\theta_0).$$

On the other hand, by comparison with $\pi^{K_1} = D_0(\overline{\rho})$ we see that θ_0 is the natural surjection

$$D_0(\overline{\rho})/D_0(\overline{\rho})_{\leq i_0} \twoheadrightarrow D_0(\overline{\rho})/D_0(\overline{\rho})_{\leq i_0'}.$$

The result follows. □